another by such smooth deformations, and we were counting the whole group as one Calabi-Yau space.) The problem is that the detailed physical properties of string vibrations, their masses and their response to forces, are very much affected by such detailed changes in shape, but once again, we have no means of selecting one possibility over any other. And no matter how many graduate students physics professors might set to work, it's just not possible to figure out the physics corresponding to an infinite list of different shapes.

This realization has led string theorists to examine the physics resulting from a sample of possible Calabi-Yau shapes. Even here, however, life is not completely smooth sailing. The approximate equations that string theorists currently use are not powerful enough to work out the resulting physics fully for any given choice of Calabi-Yau shape. They can take us a long way toward understanding, in the sense of a ballpark estimate, the properties of the string vibrations that we hope will align with the particles we observe. But precise and definitive physical conclusions, such as the mass of the electron or the strength of the weak force, require equations that are far more exact than the present approximate framework. Recall from Chapter 6—and the Price is Right example—that the "natural" energy scale of string theory is the Planck energy, and it is only through extremely delicate cancellations that string theory yields vibrational patterns with masses in the vicinity of those of the known matter and force particles. Delicate cancellations require precise calculations because even small errors have a profound impact on accuracy. As we will discuss in Chapter 12, during the mid-1990s physicists have made significant progress toward transcending the present approximate equations, although there is still far to go.

So where do we stand? Well, even with the stumbling block of having no fundamental criteria by which to select one Calabi-Yau shape over any other, as well as not having all the theoretical tools necessary to extract the observable consequences of such a choice fully, we can still ask whether any of the choices in the Calabi-Yau catalog gives rise to a world that is in even rough agreement with observation. The answer to this question is quite encouraging. Although most of the entries in the Calabi-Yau catalog yield observable consequences significantly different from our world